EVITRACE: DIGITAL EVIDENCE CHAIN-OF-CUSTODY AND INVESTIGATION TRACKING SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment of the End Semester Examination for the course

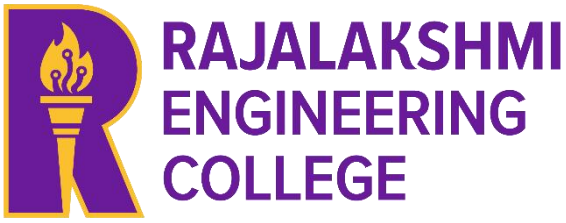
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**EVITRACE: DIGITAL EVIDENCE CHAIN-OF-CUSTODY AND INVESTIGATION TRACKING SYSTEM**" is the bonafide work of **MANISHA M 240701304**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

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DECLARATION

I, **MANISHA M 240701304** , hereby declare that the project report titled "**EVITRACE: DIGITAL EVIDENCE CHAIN-OF-CUSTODY AND INVESTIGATION TRACKING SYSTEM**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 10/11/2026

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MANISHA M

ABSTRACT

Investigation teams that handle digital incidents such as cybercrime, data theft, unauthorised access and digital fraud must be able to show where every item of digital evidence has been, who handled it and that it has not changed since collection. In many units this is still tracked through spreadsheets, paper registers, e-mail threads and separate documents, so custody gaps, unrecorded hash checks, scattered examination notes and manually chased cases are common. The problem is faced by investigation officers, evidence custodians, forensic examiners and auditors.

EviTrace was built on Zoho Creator because a single developer had to deliver a secure, role-based, workflow-driven system within one semester. The application has seven forms (Investigation Cases, Evidence Registry, Chain of Custody, Integrity Verification, Forensic Examination, Investigation Reports and Audit Trail), fourteen list, kanban, timeline and pivot reports, and an EviTrace Command Center dashboard. Deluge functions and workflows write audit entries automatically, a daily schedule e-mails officers about cases open for more than seven days, a Blueprint controls the case lifecycle and an approval process routes investigation reports to an auditor. Four role-based permission profiles and a portal viewer profile restrict access. Zoho Zia drafts investigation reports and preliminary evidence analysis, Zoho AI OCR extracts text from uploaded evidence files and a Zoho CRM connection links each case to an existing organisation and contact. AI output is stored as drafts that need human review.

The components were configured in the Creator builder and tested with sample data: the scheduled notification e-mail was received, OCR text was extracted from a sample PDF, the upload, extraction and AI analysis workflow ran end to end, and CRM organisation and contact selections were saved with a case. The result is one structured, traceable record per case and evidence item, automatic audit entries, automated reminders and AI-assisted drafting that reduces manual report writing while leaving the final judgement with the investigator.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Digital Evidence; Chain of Custody; Workflow Automation

PROJECT AT A GLANCE

Attribute	Details
Project Title	EviTrace: Digital Evidence Chain-of-Custody And Investigation Tracking System
Application Domain	Digital Forensics / Investigation and Evidence Management
Student Name & Register No.	MANISHA M – 240701304
Zoho Creator Edition	Zoho Creator (India data centre, zoho.in)
Application Name (Link Name)	EviTrace
Published / Live URL	https://manisham2024cserajala738.zohocreatorportal.in
Demo Credentials (per role)	Admin:manisha.m.2024.cse@rajalakshmi.edu.in Auditor Reviewer: manimari7765@gmail.com EviTrace Portal Viewer: manishamariappan26@gmail.com
Number of Forms / Subforms	7 / 0
Number of Reports (by type)	List: 7 | Kanban: 2 | Timeline: 1 | Pivot chart: 2 | Pivot table: 2 (total 14)
Number of Pages / Dashboards	1 (EviTrace Command Center) + customer portal
Number of Workflows	Form: 7 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Function (report action): 4 | Batch: 0
Deluge Functions / Custom Actions	7 functions / 4 custom actions
Integrations	Zoho CRM (Accounts, Contacts via OAuth connection); Zoho Zia GenAI; Zoho AI OCR
Roles & Profiles Defined	Roles: Investigation Officer, Auditor, Forensic Examiner, Evidence Custodian. Profiles: Investigation Officer, Evidence Custodian, Forensic Examiner, Auditor Reviewer, EviTrace_Portal_Viewer (portal)
Environments Used	Development → Stage → Production (Zoho Creator)
Date of Production Deployment	07 November 2026 **Production Version:** 1.1

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
GenAI	Generative Artificial Intelligence
NFR	Non-Functional Requirement
OAuth	Open Authorization
OCR	Optical Character Recognition
QR	Quick Response
RBAC	Role-Based Access Control
SHA-256	Secure Hash Algorithm, 256-bit
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Digital evidence such as laptops, mobile phones, CCTV footage, storage devices and individual files now supports a large share of investigations into cybercrime, data theft, unauthorised access and digital fraud. Unlike physical exhibits, digital evidence can be copied, altered or deleted without visible traces, so an investigation team must be able to show where each item has been, who handled it, and that its contents have not changed since collection [1], [2]. The documented record of seizure, custody, transfer, examination and disposition of an item is called the chain of custody, and the integrity of the item is commonly demonstrated by comparing cryptographic hash values such as SHA-256 taken at different points in time [3].

EviTrace is built for an investigation unit of an organisation, for example a corporate security team, a cyber-crime cell or a digital forensics laboratory. Four groups of people work together in such a unit: investigation officers who open and manage cases, evidence custodians who store evidence and record every hand-over, forensic examiners who analyse evidence and record findings, and auditors who review reports and audit activity. Each case involves one or more items of evidence, and every item passes through collection, storage, examination and reporting before the case is closed. Cases also relate to an organisation and a contact person, such as the complainant company, whose details usually already exist in a customer-relationship system.

1.2 Problem Statement

In many investigation units, case details, evidence registers, custody hand-overs, hash values, examination notes and draft reports are kept in separate spreadsheets, paper registers, e-mails and word-processor files. Because nothing links these records, the unit cannot quickly answer basic questions such as who held a particular item on a given date, whether its hash was re-checked, or which cases have remained open for too long. Good practice guides for forensic work stress that such handling must be documented and reviewable [4].

- **Fragmented records:** case, evidence, custody and examination data live in different files and cannot be traced from one to another.

- **Weak custody history:** hand-overs are recorded late, incompletely or not at all, leaving gaps in the chain of custody.
- **Unstructured integrity checks:** hash values are noted informally and the result of each re-verification is not stored against the evidence item.
- **Manual reporting without review:** investigation reports are typed from scattered notes and there is no formal review or return-for-correction step.
- **No automatic monitoring or audit trail:** stale cases are noticed only when someone remembers to check, and there is no consistent log of who did what.
- **Disconnected organisation data:** the organisation and contact linked to a case are re-typed instead of being picked from the existing CRM.

1.3 Objectives

- To design a normalised data model and forms for investigation cases, digital evidence, chain of custody, integrity verification, forensic examination, investigation reports and the audit trail.
- To automate audit logging, open-case reminders, case lifecycle control and report approval using Deluge scripting, form and schedule workflows, a Blueprint and an approval process.
- To provide role-based access, reports (list, kanban, timeline and pivot) and a Command Center dashboard for investigation officers, evidence custodians, forensic examiners, auditors and portal viewers.
- To integrate the application with Zoho CRM so that a case can be linked to an existing organisation and contact.
- To use Zoho Zia and Zoho AI OCR to assist report drafting and preliminary evidence analysis while keeping the final judgement with the investigator.
- To deploy the application securely to a production environment with governance controls such as roles, an audit trail and approval.

1.4 Scope of the Project

In scope: a web application and a portal on Zoho Creator with seven forms, fourteen reports, one dashboard page, form, schedule, approval and Blueprint workflows, seven Deluge functions, four report custom actions, role-based permission profiles (including a portal viewer profile), a Zoho CRM connection for organisations and

contacts, OCR-based text extraction from uploaded evidence files and AI-generated draft summaries.

Out of scope: forensic imaging or acquisition of devices; automatic computation of a SHA-256 hash from an uploaded file (hash values are recorded and compared as data); storage-level encryption of evidence files beyond what the platform provides; legal advice on the admissibility of evidence; two-way synchronisation of case updates back to Zoho CRM; and automatic e-mail or phone communication with CRM contacts. AI output is a preliminary draft only and is never treated as a finding of authenticity or tampering.

1.5 Justification for a Low-Code Approach

Low-code platforms let a small team build and change business applications through visual configuration, extended by a scripting layer where needed [5], [6], [7]. Zoho Creator suits EviTrace for several reasons [8]. The application is data-and-process centric, so forms, lookups, reports, kanban and timeline views come from configuration rather than code. Process control that would take real effort in a custom build, namely scheduled jobs, approval routing, a stage-based Blueprint and role-based permission profiles, is built into the platform. Hosting, backup, browser and mobile access and a customer portal are provided without separate infrastructure. Zia GenAI, OCR and the Zoho CRM integration fields are available natively, so AI assistance and CRM linkage could be added with a few lines of Deluge and a connection instead of external API plumbing. Finally, Deluge keeps the custom logic readable enough for a single developer, or a citizen developer, to maintain after the semester.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing way of working, an investigation starts when an incident is reported and the officer notes the case in a register or spreadsheet. Evidence is collected and identified only by a paper tag or a file name. Each hand-over is written on a paper form or confirmed by e-mail, and the hash value of a digital item, if calculated, is noted by hand. The examiner keeps notes in a separate document, the final report is typed in a word processor and e-mailed for review, and the supervisor periodically asks officers about pending cases. Figure 2.1 shows this flow.

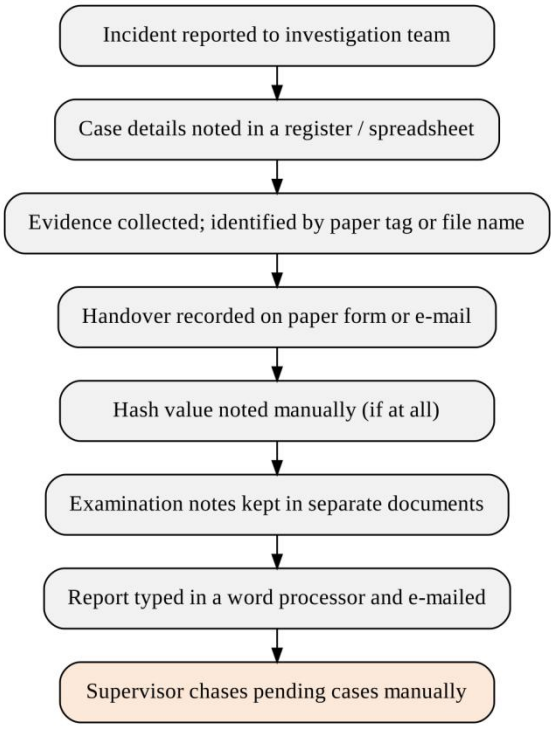


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- No single source of truth: case, evidence and custody information is spread over registers, spreadsheets and e-mails.
- The chain of custody depends on people remembering to record each transfer, and gaps are discovered only during audit or court preparation.
- Hash values and verification results are not stored in a structured way, so integrity checks cannot be searched or reported.

- Reports are prepared manually and reviewed informally, with no status, return-for-correction step or reviewer comments.
- Overdue cases are found by manual follow-up, and there is no automatic activity log.
- Organisation and contact details are typed again for each case, which invites inconsistencies.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for EviTrace itself. The traditional effort figures are estimates for a comparable custom web application, not measured values.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 8–12 weeks to build forms, role system, approval engine, scheduler, AI/OCR calls and CRM integration	3-4 weeks; forms, reports and workflows configured visually
Skills required	Front-end, back-end, database and DevOps skills	Data modelling, Creator builder and Deluge scripting
Mobile application	Separate mobile client or responsive design required	Available through the Zoho Creator mobile app and portal
Security & access control	Authentication, roles and permissions coded by hand	Built-in roles, permission profiles, portal and OAuth connections
Maintenance & change	Code change, test and redeploy for each modification	Edit forms, workflows or Deluge in the builder and republish

2.4 Literature Survey

Sources were chosen from three areas: digital-evidence handling practice, low-code research and the Zoho Creator documentation. Table 2.2 summarises them.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	E. Casey, 2011	Digital Evidence and Computer Crime, 3rd ed.	Describes handling of digital evidence, documentation and the role of integrity checks in investigations	Basis for the chain-of-custody and examination records

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	ISO/IEC, 2012	ISO/IEC 27037:2012 Guidelines for identification, collection, acquisition and preservation of digital evidence	Sets out the handling of digital evidence so that its integrity is preserved and documented	Guides the Evidence Registry, custody and integrity forms
3	A. Sahay et al., 2020	Supporting the understanding and comparison of low-code development platforms	Compares features of low-code platforms	Supports the platform choice and Table 2.1
4	A. C. Bock and U. Frank, 2021	Low-Code Platform	Defines low-code platforms and their concepts and limitations	Frames the low-code justification and limitations in Chapter 8
5	Zoho Corporation	Zoho Creator Help and Deluge reference	Documents forms, workflows, Blueprint, approvals, connections, Zia and OCR	Primary reference for every feature used

2.5 Proposed System

EviTrace replaces scattered records with seven linked forms in one Zoho Creator application. Each limitation in Section 2.2 is addressed directly: a Case Reference lookup ties evidence, examinations and reports to a case; the Chain of Custody form records every transfer with sender, receiver, reason and status; the Integrity Verification form stores the recorded and verification hashes with a result; the Investigation Reports form carries a review status and is routed through an approval process; the Audit Trail form is filled automatically by workflows; a scheduled workflow e-mails officers about cases open for more than seven days; and the CRM integration fields remove re-typing of organisation and contact details. Zia and OCR add AI-assisted drafting and preliminary analysis on top of this structured record.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner who configures the app	Manage users, roles, profiles, connections and workflows; view all data
Investigation Officer	Officer in charge of cases	Create and manage cases, assign officers, select CRM organisation and contact, coordinate the investigation
Evidence Custodian	Person responsible for storage and hand-over	Register evidence and storage details, record custody transfers, update evidence records
Forensic Examiner	Specialist who analyses evidence	Record integrity checks, examination methods and findings, run OCR and AI analysis, prepare reports
Auditor (profile: Auditor Reviewer)	Independent reviewer	Review reports, integrity results and audit trail; approve or return investigation reports
Portal Viewer (profile: EviTrace_Portal_Viewer)	External or restricted user of the portal	View authorised cases, evidence, examinations, reports and audit information

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow an officer to create an investigation case with mandatory Case Title and Opened Date, incident type, priority, assigned officer and status	Investigation Officer	High
FR-02	The system shall register digital evidence against a case with type, description, collector, date, storage location, initial hash, status and a generated QR code	Evidence Custodian	High
FR-03	The system shall record each custody transfer with sender, receiver, date and time, reason and status	Evidence Custodian	High
FR-04	The system shall store a recorded hash, a verification hash and a verification result for each integrity check	Forensic Examiner	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall record the forensic examination method, findings and status against a case and evidence item	Forensic Examiner	High
FR-06	The system shall maintain investigation reports with summary, detailed findings and approval status	Forensic Examiner, Investigation Officer	High
FR-07	The system shall add an Audit Trail entry automatically when evidence, custody, integrity, examination or report records are created	All	High
FR-08	The system shall move a case through Case Opened, Evidence Collection, Forensic Examination and Case Closed using a Blueprint	Investigation Officer	Medium
FR-09	The system shall e-mail the assigned officer daily about open cases older than seven days	Investigation Officer	Medium
FR-10	The system shall route a submitted report to an Auditor who can approve it or return it for correction	Auditor	High
FR-11	The system shall generate an AI draft report summary and detailed findings from case, evidence and examination data, and let the user apply the draft	Forensic Examiner	Medium
FR-12	The system shall extract text from an uploaded evidence file by OCR and generate a preliminary AI analysis whose review status is tracked	Forensic Examiner	Medium
FR-13	The system shall let an officer link a case to an existing Zoho CRM organisation and contact	Investigation Officer	Medium
FR-14	The system shall restrict each user to the forms, reports and pages enabled for their profile	All	High
FR-15	The system shall present kanban, timeline, pivot reports and a dashboard of cases, evidence and audit activity	All	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the Command Center shall load within a few seconds for the expected volume of a single investigation unit
NFR-02	Security	Users shall access only the forms, reports and pages enabled for their permission profile
NFR-03	Usability	The application shall be usable on web and portal, with a navigation section for each form and its reports
NFR-04	Auditability	Important investigation events shall be recorded in the Audit Trail with time and user
NFR-05	Integrity of AI output	AI output shall be stored separately as a draft and shall not be applied or treated as a finding without human action
NFR-06	Credential safety	External access to Zoho CRM shall use an OAuth connection rather than stored credentials

3.4 Hardware and Software Requirements

EviTrace runs on Zoho Creator and is scripted in Deluge. It uses Zoho CRM (Accounts and Contacts), Zoho Zia GenAI and Zoho AI OCR, and sends e-mail with the Creator sendmail task. Users need a current Chrome, Edge or Firefox browser, or the Zoho Creator mobile app (Android / iOS) or the customer portal, on a computer with about 8 GB RAM and an internet connection.

3.5 Use Case Model

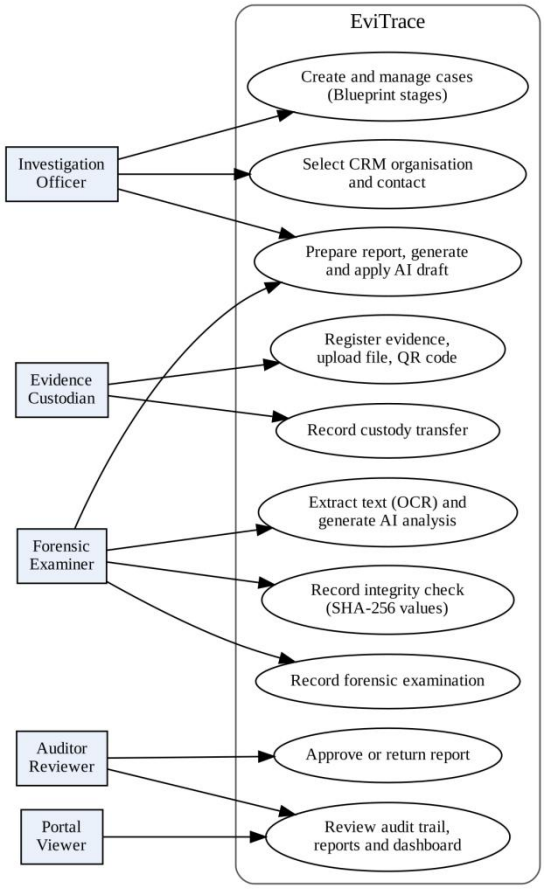


Figure 3.1: Use Case Diagram of the Application

Figure 3.1 shows the actors and their main use cases. In the two key flows, the Evidence Custodian registers evidence and records each custody transfer (FR-02, FR-03, FR-07), and the Forensic Examiner generates and applies an AI draft that the Auditor then approves or returns (FR-06, FR-10, FR-11).

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

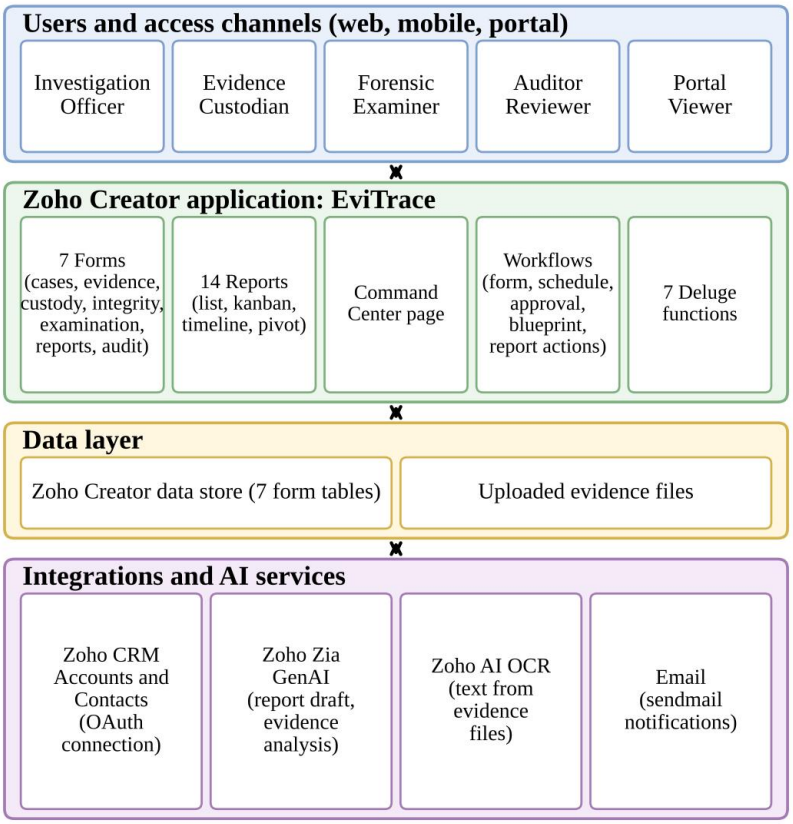


Figure 4.1: System Architecture of the Application

Figure 4.1 shows four layers. The access layer contains the four internal roles and the portal viewer, who reach the application through the web, the portal or the mobile app. The application layer is the EviTrace Zoho Creator app: seven forms, fourteen reports, the Command Center page, workflows (form, schedule, approval, Blueprint and report custom actions) and seven Deluge functions. The data layer is the Zoho Creator data store that holds the form records and the uploaded evidence files. The integration layer consists of the Zoho CRM connection (OAuth) for organisations and contacts, Zia GenAI for report drafts and evidence analysis, Zoho AI OCR for text extraction and the e-mail service used by the schedule and approval workflows.

4.2 Data Model

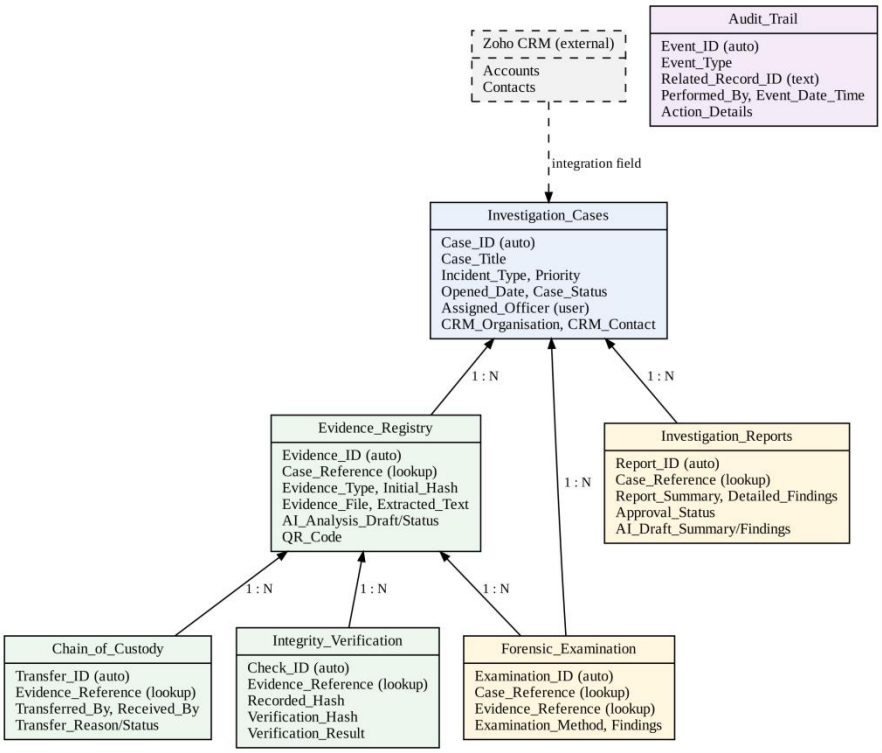


Figure 4.2: Entity–Relationship Diagram

Figure 4.2 shows the data model. Investigation_Cases is the parent form. Evidence_Registry, Forensic_Examination and Investigation_Reports each hold a Case Reference lookup to it, and Chain_of_Custody, Integrity_Verification and Forensic_Examination hold an Evidence Reference lookup to the evidence item, so one case can have many evidence items and one evidence item can have many transfers and checks. Audit_Trail stores the identifier of the affected record as text (Related Record ID) instead of a lookup, so that a single form can log events from any other form. Investigation_Cases also holds integration fields that read organisation and contact data from Zoho CRM. No subforms were needed because every repeating item (evidence, custody transfer, check) is a record in its own form so that it can have its own reports, workflows and audit entries.

4.3 Form Design

The field specifications of the seven forms are given in Table 4.1. Every form has an Auto Number identifier. Only Case Title and Opened Date are mandatory in the current build, and user-type fields such as Assigned Officer, Collected By and Examiner pick from the application users.

Table 4.1: Field Specification of the Seven Forms

Form	Fields (Link Name – Field Type – Validation / Values)
Investigation_Cases	Case_ID – Auto Number Case_Title – Single Line – **mandatory** Incident_Type – Drop Down – Cybercrime, Data Theft, Unauthorized Access, Digital Fraud, others Case_Description – Multi Line CRM_Organisation, CRM_Contact, Contacts (e-mail) – Integration fields (Zoho CRM) Priority – Drop Down – Critical, High, Medium, Low Opened_Date – Date – **mandatory** Assigned_Officer – Users (personal data) Case_Status – Drop Down – Open, In Progress, Under Review, Closed Submitted_By – Single Line – private
Evidence_Registry	Evidence_ID – Auto Number Case_Reference – Lookup (Investigation_Cases) Evidence_Type – Drop Down – Laptop, Mobile Phone, CCTV Footage, Digital File, Storage Device, Other Evidence_Description, Extracted_Text, AI_Analysis_Draft – Multi Line Evidence_File – File Upload Collected_By – Users; Collection_Date – Date-Time Storage_Location, Initial_Hash – Single Line Evidence_Status – Drop Down – Collected, In Storage, Under Examination, Released, Archived QR_Code – QR Code (encodes Evidence ID) AI_Analysis_Status – Drop Down – Not Analyzed, Draft Generated, Reviewed, Flagged for Investigation
Chain_of_Custody	Transfer_ID – Auto Number Evidence_Reference – Lookup (Evidence_Registry) Transferred_By, Received_By – Users; Transfer_Date_Time – Date-Time Transfer_Reason – Drop Down – Secure Storage, Forensic Examination, Supervisor Review, Other Transfer_Status – Drop Down – Pending Acknowledgment, Acknowledged, Rejected
Integrity_Verification	Check_ID – Auto Number Evidence_Reference – Lookup (Evidence_Registry) Checked_By – Users; Check_Date – Date-Time Recorded_Hash, Verification_Hash – Single Line Verification_Result – Drop Down – Passed, Failed, Pending Review
Forensic_Examination	Examination_ID – Auto Number Case_Reference, Evidence_Reference – Lookups Examiner – Users; Examination_Date – Date-Time Examination_Method – Drop Down – File Analysis, Metadata Analysis, Log Analysis, CCTV Analysis, Other Findings – Multi Line Examination_Status – Drop Down – Pending, In Progress, Completed, Requires Review
Investigation_Reports	Report_ID – Auto Number Case_Reference – Lookup (Investigation_Cases) Report_Summary, Detailed_Findings, Reviewer_Comments – Multi Line Submitted_By – Users Approval_Status – Drop Down – Draft, Submitted, Approved, Returned for Correction AI_Draft_Summary, AI_Draft_Findings – Multi Line
Audit_Trail	Event_ID – Auto Number Event_Type – Drop Down – Case Created, Evidence Registered, Custody Transfer, Integrity Check, Examination Updated, Report Submitted, Report Approved, Other Related_Record_ID – Single Line; Performed_By – Users Event_Date_Time – Date-Time; Action_Details – Multi Line

4.4 Process and Workflow Design

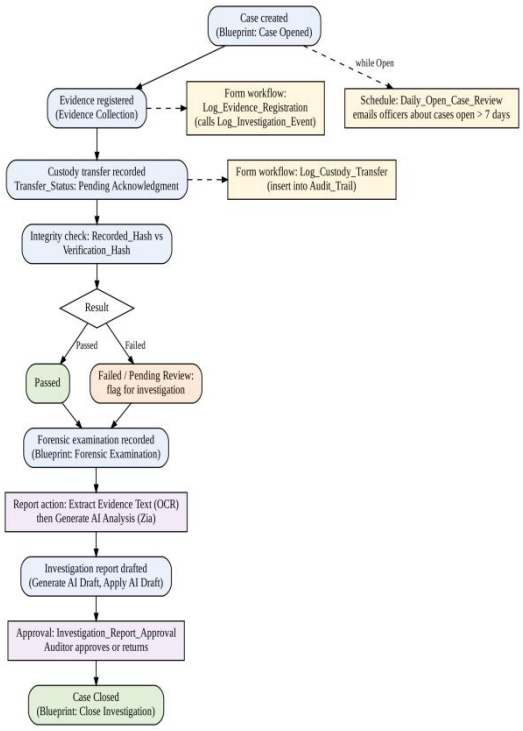


Figure 4.3: Workflow / Process Flow for the Evidence and Case Lifecycle

Figure 4.3 shows the intended path of a case. Blue boxes are user actions on forms, yellow boxes are form or schedule workflows that run automatically, and purple boxes are report actions and the approval process. A case begins at Case Opened, evidence is registered, custody transfers and integrity checks are recorded, forensic examination follows, an AI-assisted report is drafted and approved by an auditor, and the case is closed.

4.5 User Interface and Navigation Design

The navigation is organised as follows. Each form has its own section containing the form and its reports, followed by sections for Approvals, Reports and App Preferences. The EviTrace Command Center page is designed as a single-screen overview: a search box that opens a case by Case ID, a KPI panel with the number of registered cases, a gauge, charts of audit events and evidence by collection date, a donut chart of integrity checks by evidence status, a priority board and embedded case reports.

4.6 Access Control Design

Access is defined by permission profiles that are assigned to roles. Table 4.2 shows the intended permissions on each report area. V = View, E = Edit, A = Add, — = no access. Each role may edit only the records it owns in the process, and the Auditor has no edit rights.

Table 4.2: Role–Permission Matrix

Component	Investigation Officer	Evidence Custodian	Forensic Examiner	Auditor Reviewer	Portal Viewer
Investigation Cases (Board report)	A V E	V	V	V	V
Evidence Registry	V	A V E	V	V	V
Chain of Custody	V	A V	V	V	V
Integrity Verification	V	V	A V	V	V
Forensic Examination	V	V	A V E	V	V
Investigation Reports	V	V	A V	V + approve	V
Audit Trail	V	V	V	V	V
Page: EviTrace Command Center	V	V	V	V	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application EviTrace was created from scratch in Zoho Creator under the developer's workspace, with Asia/Kolkata as the time zone, the date format dd-MMM-yyyy and a 24-hour time format. The seven forms were created first, followed by the reports, the Command Center page, the workflows and the permission profiles.

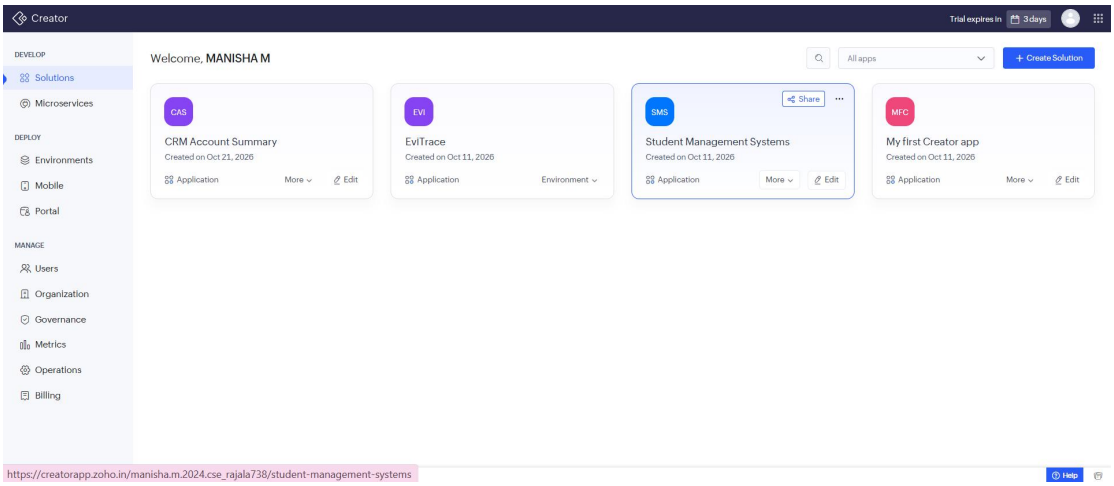


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

All seven forms use Auto Number identifiers, drop-down fields for controlled vocabularies (incident type, priority, evidence type, transfer reason and status, verification result, examination method and status, approval status) and lookup fields for relationships. User-type fields (Assigned Officer, Collected By, Transferred By, Received By, Checked By, Examiner, Submitted By, Performed By) pick from application users and are marked as personal data. Investigation_Cases is grouped into the sections Case Information, Investigation Details and Assignment and Status, and carries the CRM integration fields and the Blueprint stages. Evidence_Registry adds a File Upload field for the evidence file, a QR Code field that encodes the Evidence ID, and the AI fields Extracted Text, AI Analysis Draft and AI Analysis Status. Case Title and Opened Date are the only mandatory fields in the current build; success messages are set on each form.

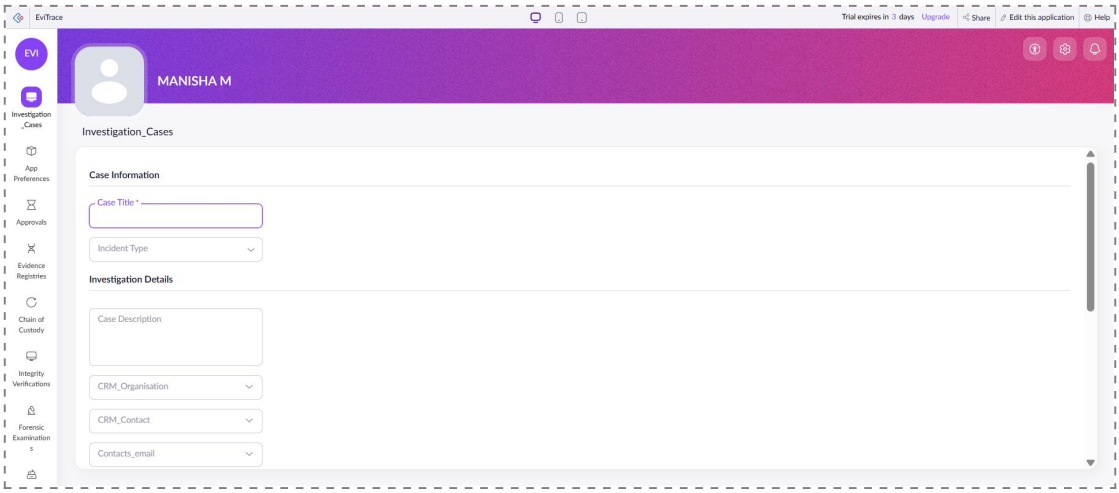


Figure 5.2: Investigation_Cases Form in the Live Mode

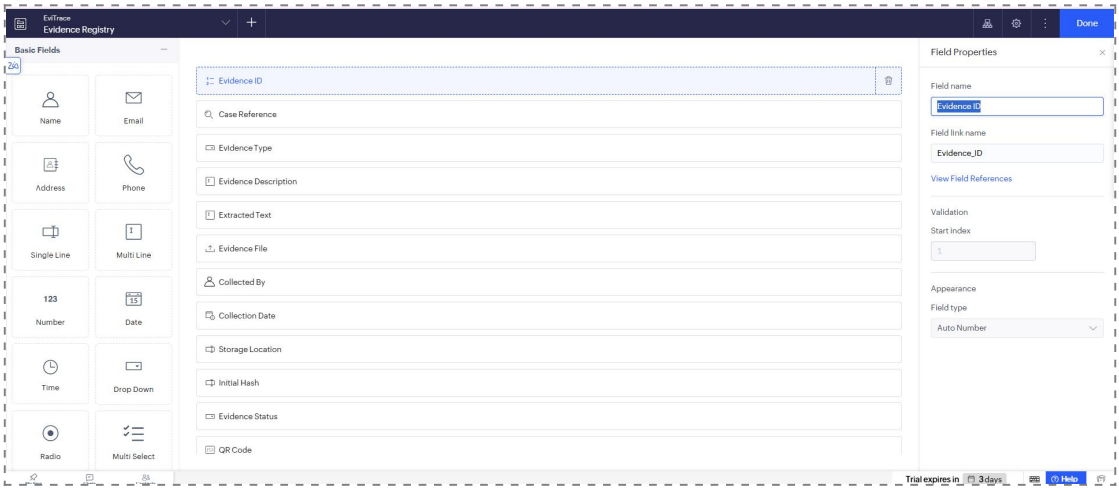


Figure 5.3: Evidence in Builder Mode

5.3 Reports

Fourteen reports were created (Table 5.1): seven default list reports (one per form), two kanban reports, one timeline report and four pivot reports. The kanban reports group cases by Case Status and custody transfers by Transfer Status, with record counts enabled. The timeline report plots audit events by Event Date & Time and Event Type. The pivot chart and pivot table reports summarise cases, integrity checks, forensic examinations and investigation reports. Four custom actions are attached to list reports: Extract Evidence Text and Generate AI Analysis on All Evidence Registries, and Generate AI Draft and Apply AI Draft on Investigation Reports Report. Each shows in the edit-record menu and asks for confirmation before it runs.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Display / Grouping	Custom Actions
Investigation_Cases Report	List	Investigation_Cases	All records incl. CRM columns	—
All Evidence Registries	List	Evidence_Registry	All records incl. QR code, extracted text	Extract Evidence Text; Generate AI Analysis
Investigation Reports Report	List	Investigation_Reports	All records incl. AI draft fields	Generate AI Draft; Apply AI Draft
Chain of Custody Report; All Integrity Verifications; All Forensic Examinations; All Audit Trails	List (4)	Respective forms	All records	—
Investigation Board Report; COC kanban	Kanban (2)	Investigation_Cases; Chain_of_Custody	Columns by Case Status / Transfer Status, record count	—
Timeline Audit Trails	Timeline	Audit_Trail	Event Type on Event Date & Time	—
Investigations Case pivot report; Integrity Verification Report	Pivot chart (2)	Cases; Integrity checks	Pivot summaries	—
Forensic Examination pivot Report; Investigations pivot report	Pivot table (2)	Examinations; Reports	Pivot summaries	—

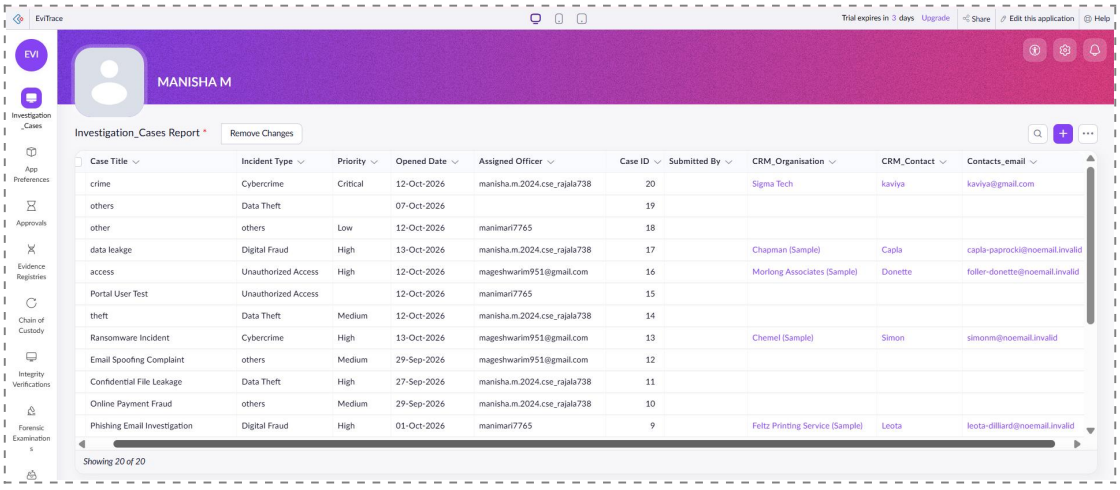


Figure 5.4: Investigation_Cases Report

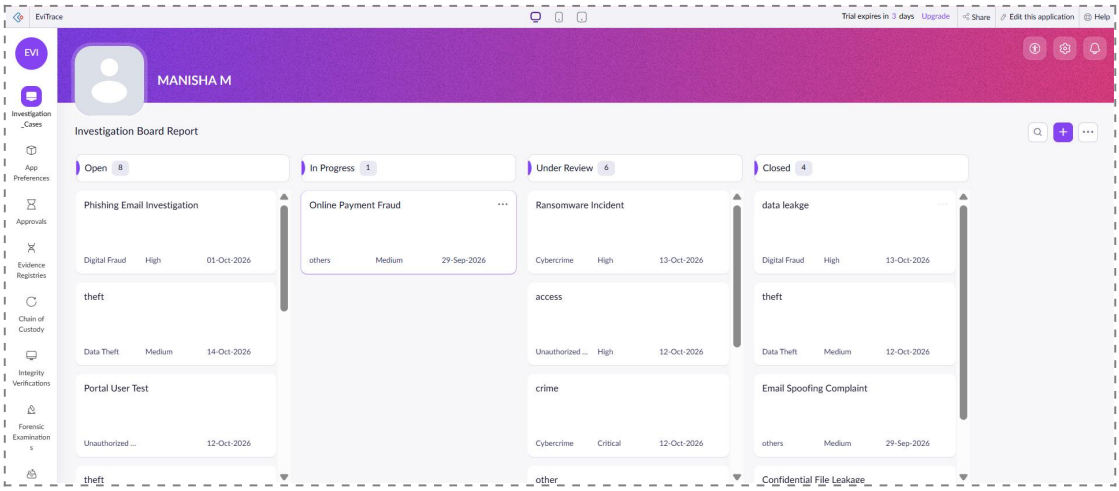
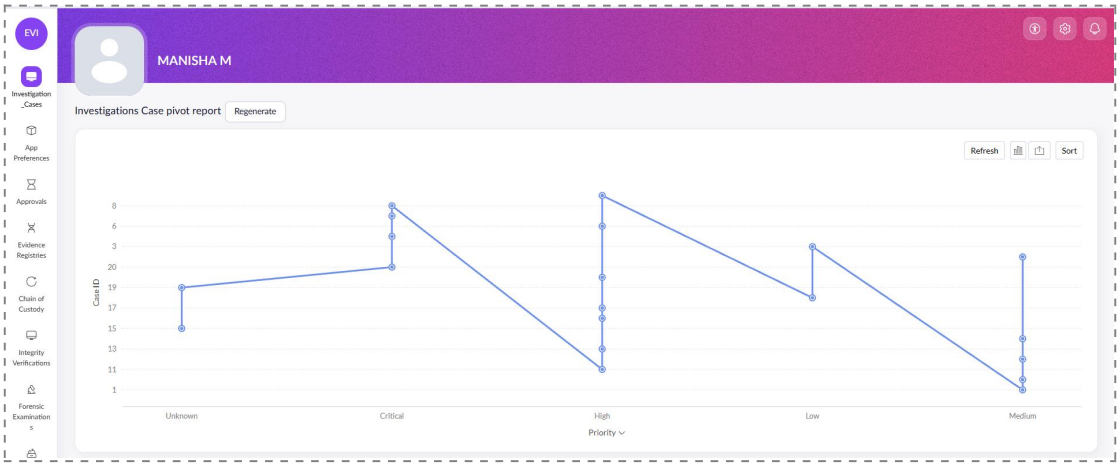


Figure 5.5: Kanban Reports



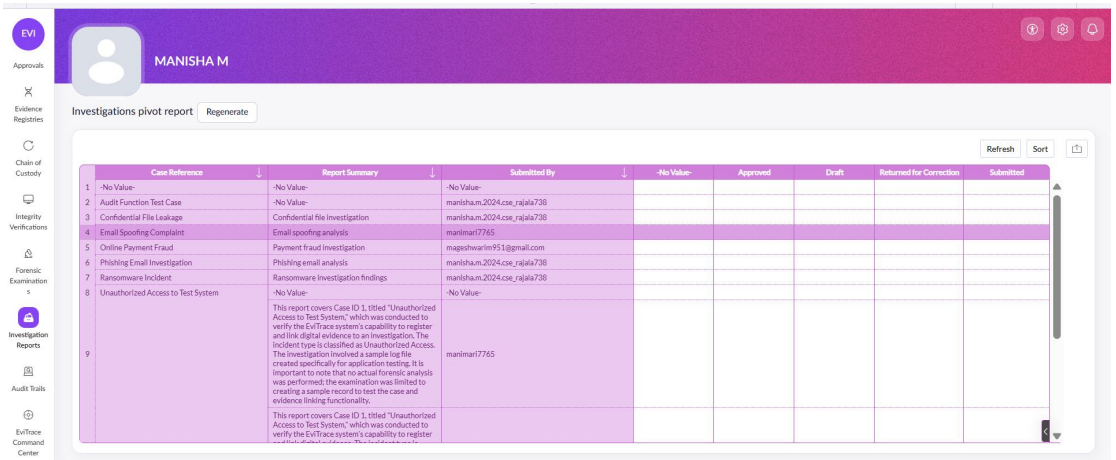


Figure 5.6: Pivot chart and Pivot Reports

5.4 Pages and Dashboards

The EviTrace Command Center page was built in the UI Builder (page layout in ZML). It contains: a search element that opens the Investigation_Cases Report filtered

by Case ID; a KPI panel with the count of registered cases; a column chart of Audit_Trail records by Action Details; a board element ranking cases grouped by Priority; a scatter chart of evidence by Collection Date; a radar (web) chart of audit events; two embedded Investigation_Cases reports; a gauge showing the case count; a line chart of audit events by Event Type; and a donut chart of integrity checks grouped by the evidence status of the referenced evidence. The page is enabled as a tab in every role profile.

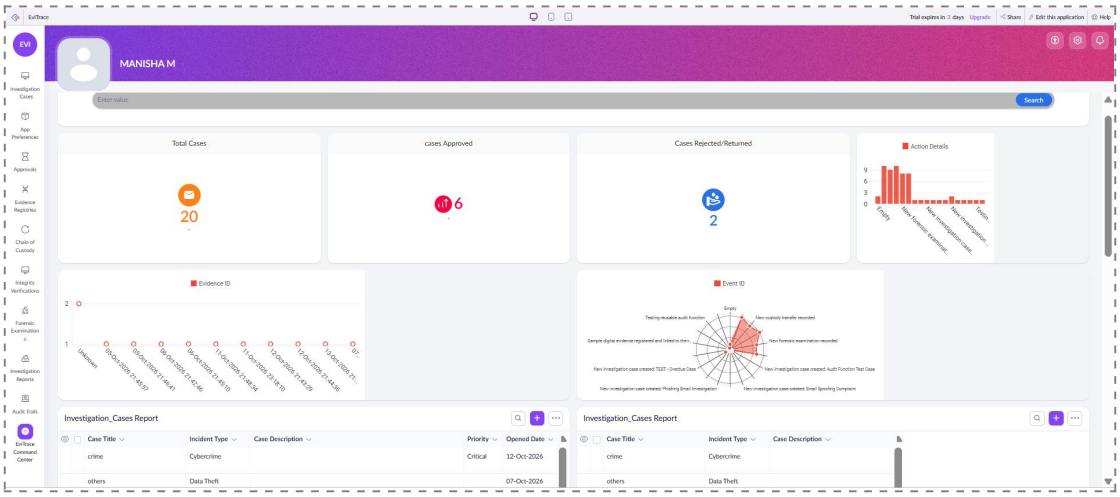


Figure 5.7: EviTrace Command Center Page

5.5 Deluge Scripting

Deluge [9] is used in seven functions, in the actions of the form workflows, in the schedule and in the approval process. The main functions and workflow scripts are listed in Appendix C; the AI prompt-building and OCR functions are described in Section 5.9.

Code 5.1: Log_Investigation_Event function (DS-06)

```
void Log_Investigation_Event(string eventType, string relatedRecordID,
string performedBy, string actionDetails)
{
    insert into Audit_Trail
    [
        Event_Type=eventType
        Related_Record_ID=relatedRecordID
        Performed_By=performedBy
        Event_Date_Time=zoho.currenttime
        Action_Details=actionDetails
        Added_User=zoho.loginuser
    ]
}
```

Code 5.2: Daily_Open_Case_Review schedule (DS-10)

```
// DS-10 | Schedule: Daily_Open_Case_Review | Daily, Asia/Kolkata
overdueDate = zoho.currentdate.addDay(-7);
overdueCases = Investigation_Cases[Opened_Date < overdueDate &&
Case_Status == "Open"];
for each  caseRec in overdueCases
{
    info "Overdue Case: " + caseRec.Case_Title;
    officerEmail = caseRec.Assigned_Officer.email;
    if(officerEmail != null && officerEmail != "")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :officerEmail
            subject :"EviTrace - Overdue Investigation Case"
            message :"An investigation case assigned to you has been open
for more than 7 days..."
        ]
    }
    else
    {
        info "No assigned officer email found for this case.";
    }
}
```

How the scripts work. DS-06 is the single entry point for audit logging: it stamps the event with the current time and the logged-in user. DS-08 to DS-09 run when a record is added to the evidence, custody, integrity, examination and report forms and write the matching Audit_Trail row. DS-10 computes the cut-off date as today minus seven days, fetches Open cases opened before it with one criteria fetch, and sends one e-mail per case when the assigned officer has an e-mail address; the `info` statements were used to check which cases were picked. DS-01 builds a plain-text input from the case, its evidence and its examinations and sends it to Zia with instructions to use only the supplied information; DS-02 then checks that both marker headings are present before saving, so a malformed AI response never overwrites a draft. DS-04 stops with a readable error message if there is no file or no text. DS-05 refuses to run unless text has been extracted. In all functions, errors are returned as strings that the custom action shows to the user.

Result observed. The scheduled review sent a notification e-mail to the assigned officer when tested; OCR extracted readable text from a sample PDF; and the sequence upload file → extract text → generate analysis saved a draft and set the status to Draft Generated.

5.6 Workflows and Automation

Table 5.2 lists every workflow. Form workflows write the audit trail, the schedule monitors old cases, the approval process controls report review, the Blueprint controls the case lifecycle and function workflows attached to report custom actions run the AI and OCR functions for one selected record at a time.

Table 5.2: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action(s)	Script Ref.
Log_Evidence_Registration; Log_Custody_Transfer; Log_Integrity_Check; Log_Forensic_Examination; Log_Investigation_Report	Form (5, active)	On Add → On Success of the respective form	Insert Audit_Trail row (directly or via Log_Investigation_Event)	DS-06, 08, 09
Log_Case_Creation; Auto_Set_Submitted_By	Form (2, inactive)	Investigation_Cases – On Add → On Success	Audit entry; read portal user e-mail	DS-12
Daily_Open_Case_Review	Schedule	Daily, Asia/Kolkata; Open cases older than 7 days	E-mail the assigned officer	DS-10
Investigation_Report_Approval	Approval	Report with Approval Status = Submitted; Level 1 role Auditor	Approve: e-mail, status Approved. Reject: e-mail, status Returned for Correction	DS-11
Investigation_Case_Lifecycle	Blueprint	Stage transitions on Investigation_Cases	Start Evidence Collection; Begin Forensic Examination; Close Investigation	—
Extract_Evidence_Text_rep; Generate_AI_Analysis	Function (report action)	Custom actions on All Evidence Registries	Run DS-04; run DS-05 on the selected record	DS-04, 05
Generate_AI_Report_Workflow; Apply_AI_Report_Workflow	Function (report action)	Custom actions on Investigation Reports Report	Run DS-02; run DS-03 on the selected record	DS-02, 03

Blueprint. Investigation_Case_Lifecycle has four stages (Case Opened, Evidence Collection, Forensic Examination, Case Closed) and three transitions: Start Evidence Collection, Begin Forensic Examination and Close Investigation. Because the Blueprint controls movement, a case cannot jump from Case Opened directly to Case Closed. **Approval.** The approval process applies to Investigation_Reports records whose Approval Status is Submitted and has one level whose approver is the Auditor role. **Batch workflows** were not required, because the report custom actions process one selected record at a time. One additional function workflow,

Extract_Evidence_Text_fun, was created while experimenting and has no action attached; the working version is Extract_Evidence_Text_rep.

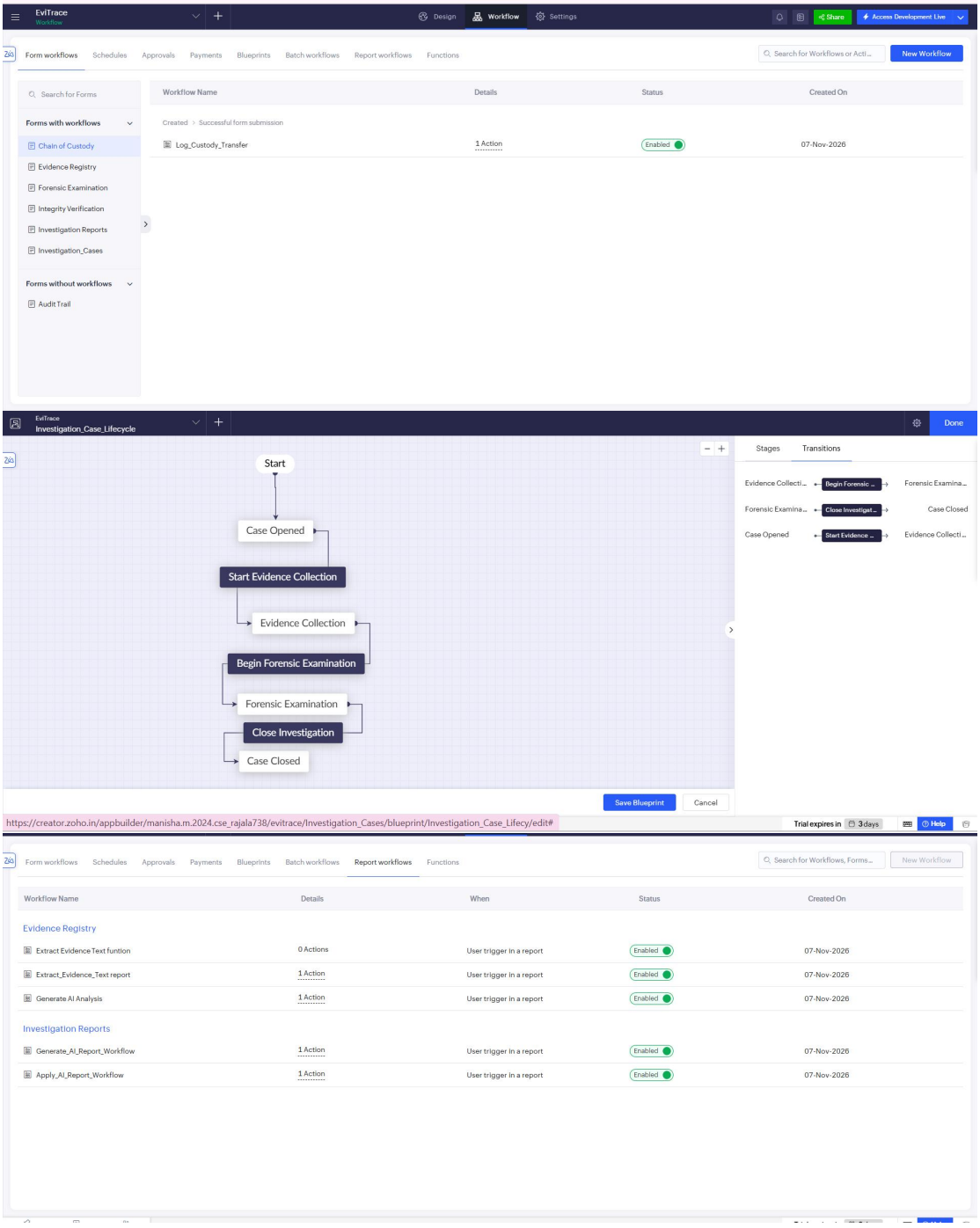


Figure 5.8: Workflows, Blueprint and report workflow

5.7 Integrations

Zoho CRM is integrated through a connection named zoho_crm_connection (Zoho CRM connector, OAuth) with read and search scopes for accounts and deals records. Two data sources, zoho_crm_accounts and zoho_crm_contacts, expose the CRM

Accounts and Contacts modules to Creator. The Investigation_Cases form uses them as integration fields: CRM Organisation (display: account name), CRM Contact (display: first name) and a contact e-mail field. In tests with the existing CRM records Sigma Tech and Kaviya M, both lists were populated from CRM, both selections were saved with the case and remained visible after submission. The integration links CRM records to cases; it does not push case updates back to CRM and does not send e-mails or calls to contacts.

Zia GenAI and Zoho AI OCR are used from Deluge, with no separate API keys: the `zia` task and `zoho.ai.recognizeText` run in the context of the Zoho account (Table 5.3).

Table 5.3: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
CRM link	Zoho CRM – Accounts and Contacts	Connection + integration (data source) fields	OAuth 2.0 via Zoho connection	Read organisation and contact for a case
AI report and analysis	Zoho Zia GenAI	`zia` task in Deluge (temperature 0.1)	Zoho account context	Case or evidence text in; draft text out
Text extraction	Zoho AI OCR	`zoho.ai.recognizeText(file,"OCR","eng")`	Zoho account context	Evidence file in; extracted text out
E-mail	Zoho Creator mail	`sendmail`	Zoho account	Overdue case and approval notifications

Code 5.3: AI service calls (DS-04 and DS-05 excerpts)

```
// DS-04 | OCR on the uploaded evidence file
ocrResponse =
zoho.ai.recognizeText(evidenceRecord.Evidence_File,"OCR","eng");
extractedText = ifnull(ocrResponse.get("text"),"");

// DS-05 | Preliminary AI analysis of the extracted text
aiResponse = zia
[
    message:prompt
    context:"You assist with preliminary digital evidence analysis.
Evidence integrity and factual accuracy are essential."
    parameters:{"temperature":0.1}
];
evidenceRecord.AI_Analysis_Draft = aiResponse.toString();
evidenceRecord.AI_Analysis_Status = "Draft Generated";
```

No external REST API (invokeurl), webhook or Creator REST API was needed, so no Postman test was performed; the integration evidence is the connection setup and the saved CRM selections.

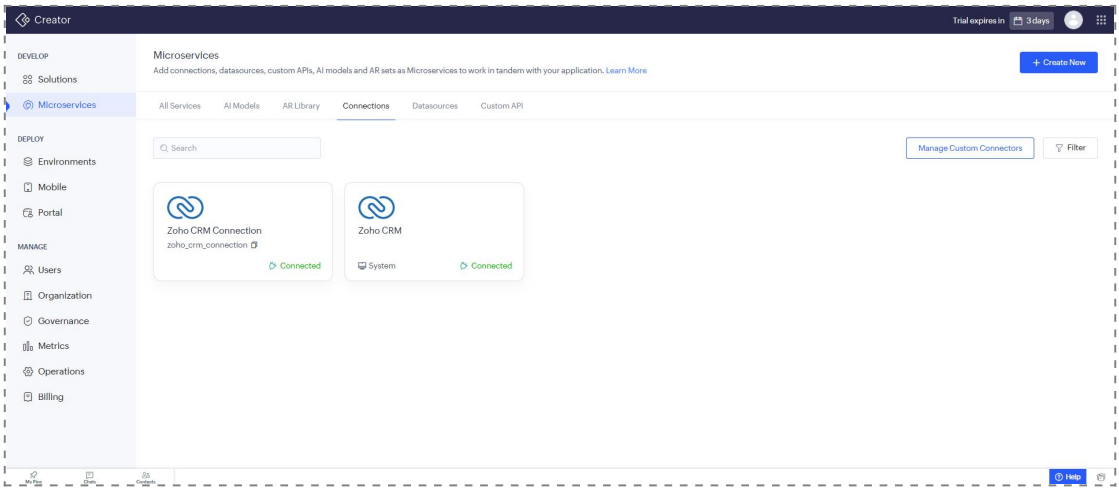


Figure 5.9: Zoho CRM Connection

5.8 User Management and Role-Based Access

Four roles sit under the top-level role in the role hierarchy: Investigation Officer, Auditor, Forensic Examiner and Evidence Custodian. Permission profiles control what each role can see and edit (Table 4.2). In every internal profile, all seven forms and the Command Center page are enabled as tabs and the report for each form is viewable; edit rights are granted only on the report that the role owns. A separate customer-portal profile, EviTrace_Portal_Viewer, gives portal users a read-only view of the same reports.

The Auditor role is also the approver in Investigation_Report_Approval. In the current build, the module-level Create permission is enabled in every profile; tightening this for the Auditor and Portal Viewer is listed under future enhancements. Record-level criteria were not configured.

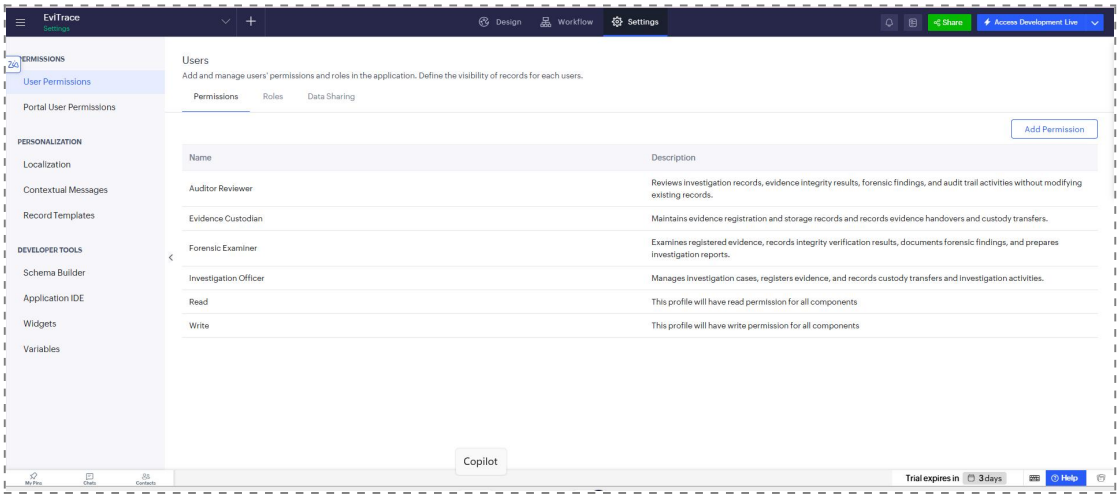


Figure 5.10: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

Zia GenAI report assistant. DS-01 gathers the case title, incident type, description, the type and description of each evidence item and the method and findings of each examination, and sends them to Zia with instructions to use only the supplied information, not to invent evidence, dates or conclusions, to flag sample or unverified data, to say when information is missing and to return two sections: REPORT SUMMARY and DETAILED FINDINGS.

OCR and AI evidence analysis. The Extract Evidence Text action uses `zoho.ai.recognizeText` to read an uploaded evidence file, and Generate AI Analysis sends the text and evidence metadata to Zia. The prompt asks the model to separate observations from possibilities, list missing information and never state that evidence is genuine or tampered with. The AI Analysis Status field (Not Analyzed, Draft Generated, Reviewed, Flagged for Investigation) records the human review state.

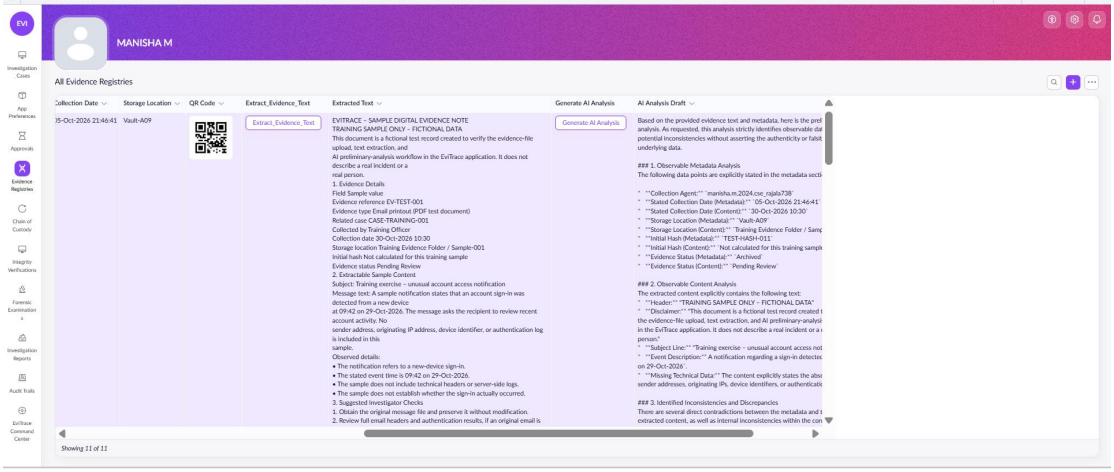


Figure 5.11: OCR Text and AI Analysis for an Evidence Record

5.10 Pro-Code Extension

No pro-code extension (Catalyst, JavaScript widget or Creator REST API) was used, because audit logging, reminders, AI calls and OCR were achievable with Deluge functions and workflows. The main gap is that hashes are typed in by the examiner; a Catalyst function could compute the SHA-256 hash of an uploaded file automatically.

5.11 Mobile and Portal Access

A customer portal lets restricted users view authorised information through the EviTrace_Portal_Viewer profile, and internal users can use the Zoho Creator mobile app.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done at four levels. Unit testing exercised each form with valid and invalid input and each Deluge function by running it from its custom action. Integration testing covered the workflows (audit logging, schedule, approval, Blueprint), the OCR and Zia calls and the CRM integration fields. Role-based testing used a login for each permission profile to confirm the visible tabs and actions. User acceptance testing collected feedback from users acting in their roles. Results of tests that have been executed are marked in Table 6.1; cells shown in yellow are still to be filled from the final test run.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create a case with all mandatory fields	Enter Case Title and Opened Date; submit	Record created; success message	Case created successfully	Pass
TC-02	FR-02	Register evidence for a case	Select case, type, hash, location; submit	Record saved; QR code generated	Evidence saved and QR code generated	Pass
TC-03	FR-03	Record a custody transfer	Select evidence, sender, receiver, reason; submit	Transfer saved with status	Custody transfer recorded successfully	Pass
TC-04	FR-04	Record an integrity check	Enter recorded and verification hash; result Passed	Record saved	Integrity check completed successfully	Pass
TC-05	FR-05	Record a forensic examination	Select case, evidence, method, findings	Examination saved	Forensic examination saved successfully	Pass
TC-06	FR-08	Blueprint stage progression	Run Start Evidence Collection on a new case	Case moves to Evidence Collection	Case moved to Evidence Collection stage	Pass
TC-07	FR-09	Daily open-case review	Run schedule with a case open more than 7 days	E-mail sent to assigned officer	Notification e-mail received	Pass
TC-08	FR-11	Generate and apply AI draft	Run Generate AI Draft, then Apply AI Draft	Draft saved, then copied to report fields	AI draft generated and applied successfully	Pass
TC-09	FR-12	OCR on a sample PDF	Upload PDF; run Extract Evidence Text	Text saved in Extracted Text	Text extracted successfully	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-13	AI evidence analysis end to end	Upload, extract, generate analysis	Analysis saved; status Draft Generated	Full workflow completed	Pass
TC-11	FR-14	CRM organisation and contact	Select Sigma Tech and Kaviya M; save	Values saved and visible after submit	Selections saved and visible	Pass

6.3 Debugging Deluge Scripts

Three techniques were used. First, `info` statements in the schedule (for example the case title, case ID and a message when no officer e-mail exists) showed which cases the criteria had selected. Second, every function returns a clear message string for each failure path (record not found, no file, no readable text, incomplete draft), so errors appear as messages when a custom action is run. Third, the Deluge editor's execute option and the workflow execution logs were used to check function arguments and results. The main real problem concerned portal submission: the workflow that logged case creation and the workflow that read the portal user's e-mail did not work correctly when a case was submitted through the portal during testing. Both were disabled so that case submission is not blocked, and the reusable Log_Investigation_Event function was kept.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Submitting a case through the portal failed when the case-creation logging workflow was active	Workflow action ran in the portal-user submission context	Log_Case_Creation and Auto_Set_Submitted_By disabled; function retained	Closed (workaround)
BUG-02	Duplicate function workflow Extract_Evidence_Text_fun had no action	Created while experimenting with the report action	Working workflow Extract_Evidence_Text_rep used for the custom action	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Portal Viewer	Portal User	Restricted access to records is useful.	Configured portal permissions

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

EviTrace was developed using the Zoho Creator Development environment, where forms, reports, workflows, Deluge functions and user permissions were configured and tested. The application was configured across Development, Stage and Production environments using Version 1.0 on 07 November 2026. The Production environment provides access to the application, while the EviTrace portal enables email-based authentication and controlled access through Admin, Auditor Reviewer and Portal Viewer permissions. The portal was also tested in an incognito browser to verify external access. Version management in Zoho Creator supports tracking application changes and maintaining previous versions for future updates.

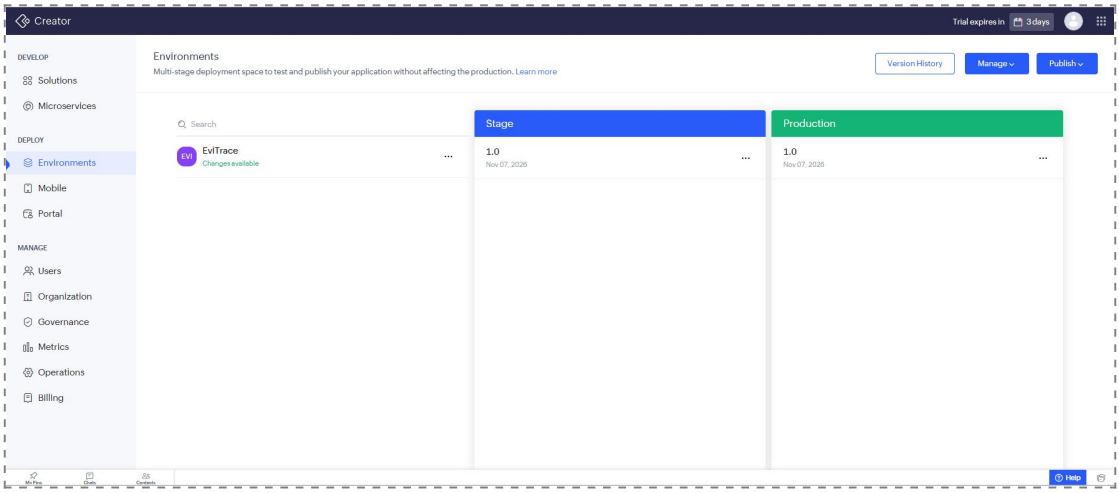


Figure 7.1: Environments / Publish to Production

7.2 Performance Optimisation

Performance work focused on the Deluge code, because the volume of data in a single investigation unit is modest. Table 7.1 lists the techniques applied. The effect was not measured with timings in this project, so the improvement column is qualitative.

Table 7.1: Performance Optimisation Applied

Optimisation Applied	Component	Approach Used	Effect
Filter in the fetch criteria instead of looping over all records	DS-10 (schedule)	Single fetch with Opened_Date < cut-off and Case_Status == "Open"	Only overdue open cases are loaded

Optimisation Applied	Component	Approach Used	Effect
Validate before expensive calls	DS-04, DS-05	Return early when the record, file or text is missing	No OCR or Zia call is wasted
Check the AI response before saving	DS-02, DS-03	Both section markers verified; both draft parts required	Avoids rework from partial drafts
Reuse one logging function	DS-06	Common Log_Investigation_Event function	Less duplicated code

7.3 Security Features

Table 7.2: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permission profiles	Four internal profiles and one portal profile	Unauthorised viewing and editing
Edit rights limited to the owning role	Evidence, examination and case reports	Accidental or unauthorised change of another role's records
Personal-data flag on user fields	Assigned Officer, Collected By, Transferred By, Received By, Examiner, Performed By	Exposure of personal data
OAuth connection with limited scopes	zoho_crm_connection (read and search)	Credential leakage and excessive CRM access
Approval before a report is treated as approved	Investigation_Report_Approval	Unreviewed reports
Audit trail of key events	Audit_Trail form and workflows	Untraceable actions
AI guardrails in prompts and human review	DS-01, DS-05; draft fields and status	Fabricated or over-confident AI conclusions

7.4 Governance and Administration

The Audit_Trail form and the Timeline Audit Trails report give the main governance record: each audit row stores the event type, the related record, the user, the time and a short description. Naming follows one pattern across the application (forms in singular business names, link names such as Case_Reference and Evidence_Reference, reports prefixed by their purpose, workflows prefixed Log_ for logging). Data can be exported from reports, and the platform backs up the hosted data. User licences and roles are managed in the Users & Control area by the application owner, who is also responsible for maintaining connections, schedules and profiles. Known gaps to be maintained are listed in Section 8.4.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario below follows one case, a suspected data-theft incident at the organisation Sigma Tech whose contact is Kaviya M, through the roles.

Step 1. The Investigation Officer creates the case, picks Sigma Tech and Kaviya M from Zoho CRM, sets the priority and the assigned officer. **Step 2.** The Evidence Custodian registers a laptop as evidence with its initial hash and storage location, and the audit entry appears. **Step 3.** The custodian records the transfer of the laptop to the Forensic Examiner. **Step 4.** The Forensic Examiner records an integrity check and the examination, and uploads a document for OCR and AI analysis. **Step 5.** The examiner generates and applies the AI report draft and submits the report. **Step 6.** The Auditor approves the report and the dashboard and audit timeline reflect the activity.

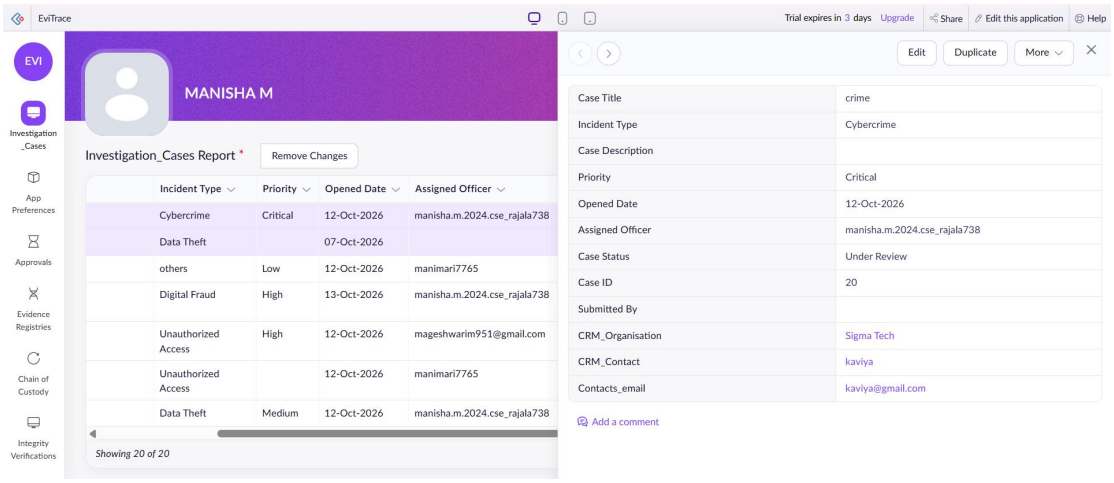


Figure 8.1 Case created with CRM organisation and contact

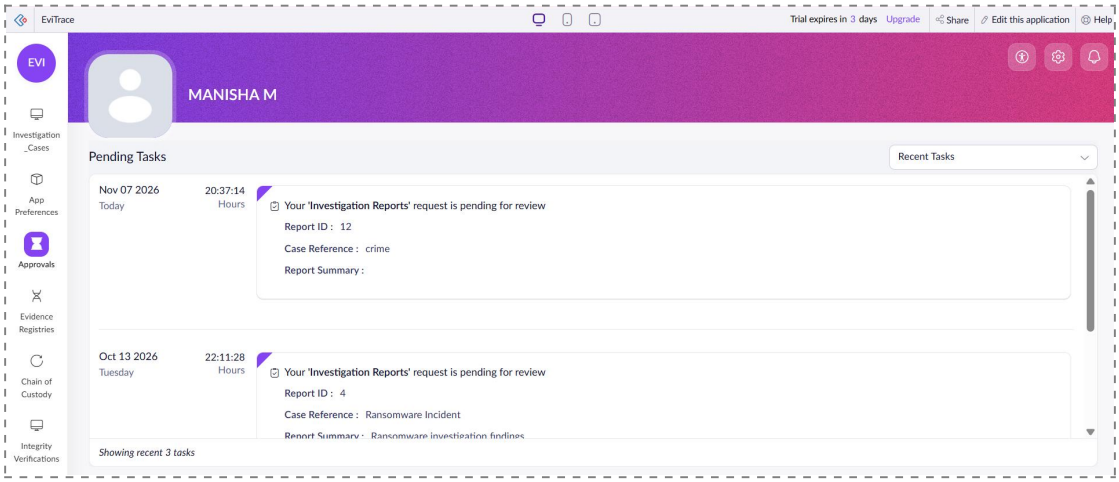


Figure 8.2: Pending Tasks for Auditor Approval

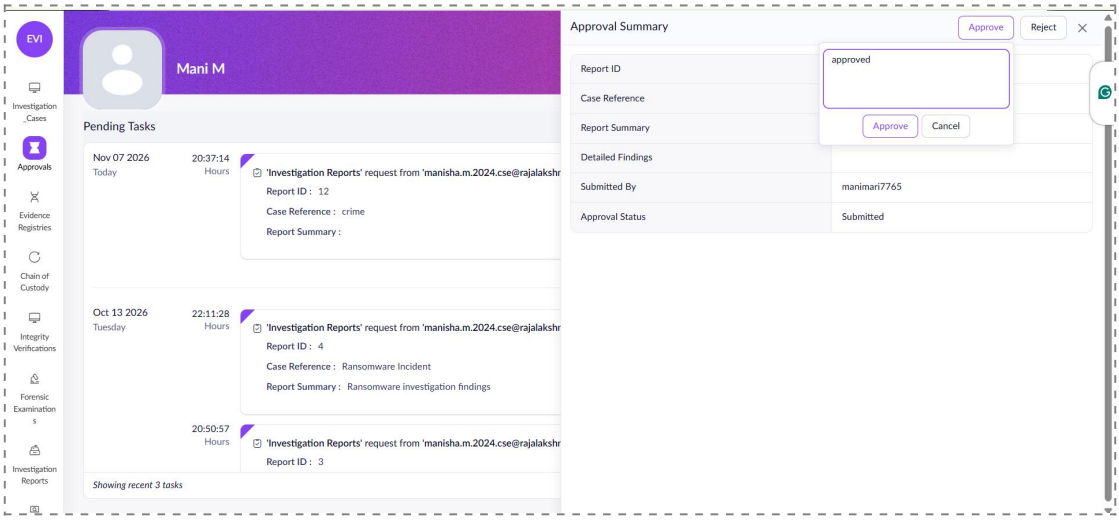


Figure 8.3: Auditor approval and updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Data model and forms for investigation records	Sections 4.2, 4.3, 5.2	Seven linked forms with lookups, Auto Number IDs and controlled drop-downs
2. Automation using Deluge, workflows, Blueprint, approval	Sections 5.5, 5.6	Automatic audit entries, daily open-case e-mail (tested), Blueprint lifecycle, auditor approval
3. Role-based reports and dashboard	Sections 5.3, 5.4, 5.8	Fourteen reports, Command Center page, four role profiles and a portal viewer profile
4. Zoho CRM integration	Section 5.7	Organisation and contact selected from CRM and saved with the case (tested)

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
5. AI-assisted drafting with human review	Sections 5.9	OCR and AI analysis tested end to end; AI output stored as drafts with review status
6. Secure deployment and governance	Chapter 7	Profiles, OAuth connection, audit trail, approval; deployment details in Section 7.1

8.3 Comparison with the Existing System

The comparison is qualitative because no timing study of the manual process was carried out. Case and evidence records move from registers and spreadsheets to linked forms with lookups, so each item has one traceable record. Custody history moves from paper forms or e-mail to a Chain of Custody form with status, a kanban view and audit entries. Report preparation moves from manual typing to an AI draft, an apply step and an approval, which reduces drafting effort while keeping human review. Old open cases, which nobody followed up before, now trigger a daily e-mail, and every key event leaves an Audit_Trail row that was not kept earlier. Access is role-based from the web, portal and mobile app instead of local files.

8.4 Limitations of the Low-Code Approach

Several constraints appeared during the project. Page and dashboard styling is limited to the options of the UI Builder, so the Command Center follows the platform's component set. The AI and OCR features depend on the Zia and OCR services of the platform, and their output quality, limits and availability are outside the developer's control. Hash values must be entered because the application does not compute a SHA-256 hash from an uploaded file, and the verification result is selected by the examiner instead of being derived by comparing the two hash fields. The portal submission problem with the case-creation logging workflow (BUG-01) showed that workflows behave differently for portal users, and the workaround was to disable them. The Create permission is enabled at module level in every profile, which is wider than the intended matrix in Table 4.2 for the Auditor and the Portal Viewer. The CRM integration links data in one direction only. Finally, the application depends on Zoho Creator, so moving it to another platform would require rebuilding the forms and Deluge logic.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of EviTrace is no-code configuration: forms, lookups, drop-downs, reports, kanban, timeline and pivot views, the dashboard, roles and profiles, the Blueprint and the approval process. Low-code Deluge was needed for the audit logging functions, the schedule, the Zia prompts and the OCR call, and for string handling of the AI response. Pro-code would be needed for the next level of capability: computing SHA-256 hashes from uploaded evidence files, verifying them automatically and perhaps tamper-evident storage. The point to move to pro-code is when a requirement depends on processing that Deluge or the platform components cannot express safely, as with hashing large evidence files.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

EviTrace is a Zoho Creator application for managing investigation cases and digital evidence from collection to closure. It combines seven linked forms, fourteen reports, a Command Center dashboard, audit-logging workflows, a daily open-case review, a Blueprint for the case lifecycle, an auditor approval process, role-based access with a portal profile, Zoho CRM integration and AI-assisted report drafting and evidence analysis using Zia and OCR.

The project shows that a structured, traceable evidence workflow can be built on a low-code platform by one developer within a semester, with Deluge used only where configuration was not enough. The AI features were designed as drafts for human review, and the report is explicit that SHA-256 comparison supports integrity checking but does not by itself prove the origin or authenticity of evidence.

9.2 Future Enhancements

- Compute the SHA-256 hash of uploaded evidence files automatically (for example with a Zoho Catalyst function) and derive the verification result by comparing hashes.
- Restrict the Create permission for the Auditor and Portal Viewer profiles, add record-level criteria for portal users and re-enable case-creation logging after fixing the portal issue.

9.3 Learning Outcomes and Career Relevance

Building EviTrace gave practical experience in data modelling with lookups, designing role-based access, writing Deluge functions, configuring schedule, approval and Blueprint workflows, connecting Zoho CRM through OAuth and using Zia and OCR responsibly. These skills relate directly to roles such as low-code developer, solutions consultant and automation engineer, and the experience of limiting AI output to reviewable drafts is relevant to any role that introduces AI into a business process.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Space menu with one section per form	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Auto Number, Single/Multi Line, Drop Down, Date, Date-Time, Lookup, Users, File Upload, QR Code, Integration	5.2	Figure 5.3
II	Lookup field (relationship between forms)	Y	Case_Reference, Evidence_Reference	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	N	Not used; child records are separate forms	4.2	—
II	Field validations / layout rules	Y	Mandatory fields only (Case Title, Opened Date)	5.2	Figure 5.2
II	List report with filter, grouping, custom action	Y	All Evidence Registries; Investigation Reports Report	5.3	Figure 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Two kanban reports and one timeline (no calendar or map)	5.3	Figure 5.5
II	Pivot table / chart report	Y	Two pivot charts and two pivot tables	5.3	Figure 5.6
II	Page / dashboard built with UI Builder	Y	EviTrace Command Center	5.4	Figure 5.7
II	Deluge – form events (validate / success / field actions)	Y	On Add → On Success workflows (no validate script)	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	Y	Seven functions with fetch and for each	5.5	Code 5.3
II	Email / in-app notifications	Y	Schedule e-mail; approval e-mails	5.5, 5.6	Code 5.2
II	Form workflow	Y	Log_* workflows	5.6	Figure 5.8
II	Scheduled workflow	Y	Daily_Open_Case_Review	5.6	Figure 5.8
II	Approval process	Y	Investigation_Report_Approval	5.6	Figure 5.8
II	Blueprint	Y	Investigation_Case_Lifecycle	5.6	Figure 5.8
II	Batch / report workflow	Partial	Report custom actions (function workflows); no batch workflow	5.6	Figure 5.4
II	Testing & debugging evidence	Y	Test cases, info output, defect log	6	Figure 5.8
III	Integration with a Zoho service	Y	Zoho CRM via zoho_crm_connection	5.7	Figure 5.9
III	Connection / REST API call (invokeurl)	Y	Connection used for CRM data sources; no invokeurl call	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Four roles, four internal profiles, one portal profile	5.8	Figure 5.10
III	Record-level access criteria / customer portal	Partial	Portal profile configured; no record-level criteria	5.8, 5.11	Figure 5.10

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Zia / AI feature	Y	Zia GenAI drafts and analysis; OCR	5.9	Figure 5.11
III	Analytics (charts / Zoho Analytics)	Y	Charts and gauge on the Command Center; pivots	5.4, 5.9	Figure 5.7
IV	Performance optimisation applied & measured	Partial	Applied; not measured with timings	7.2	Table 7.1
IV	Deployment across environments	Y	Deploy,envrironments	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Profiles, OAuth connection, private/personal-data fields	7.3	Table 7.2
IV	Governance (audit trail, administration)	Y	Audit_Trail form and timeline	7.4	Figure 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used; justified in 5.10	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	EviTrace_Portal_Viewer profile	5.11	Figure 5.10

Total features fully implemented: 26 / 32 (3 partially implemented).

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – DELUGE SOURCE CODE

C.1 Generate_Report_Draft

```
// Script ID : DS-02
// Location  : Function
// Purpose   : Generate and split the AI draft for a report

string Generate_Report_Draft(int reportID)
{
    reportRecord = Investigation_Reports[Report_ID == reportID];
    if(reportRecord.ID == null)
    {
        return "Error: Investigation report not found.";
    }
    if(reportRecord.Case_Reference == null)
    {
        return "Error: No case is linked to this report.";
    }
    caseID = reportRecord.Case_Reference.Case_ID;
    aiDraft = thisapp.AI_Report_Assistant(caseID);
    summaryMarker = "# REPORT SUMMARY";
    findingsMarker = "# DETAILED FINDINGS";
    if(!aiDraft.contains(summaryMarker)
|| !aiDraft.contains(findingsMarker))
    {
        return "Error: AI response does not contain both report sections.
Draft was not saved.";
    }
    summaryStart = aiDraft.indexOf(summaryMarker) +
summaryMarker.length();
    findingsStart = aiDraft.indexOf(findingsMarker);
    summaryText = aiDraft.subString(summaryStart,findingsStart).trim();
    findingsText = aiDraft.subString(findingsStart +
findingsMarker.length(),aiDraft.length()).trim();
    reportRecord.AI_Draft_Summary=summaryText;
    reportRecord.AI_Draft_Findings=findingsText;
    return "AI draft generated successfully for Report ID: " +
reportRecord.Report_ID;
}
```

C.2 Apply_AI_Report_Draft

```
// Script ID : DS-03
// Location  : Function
// Purpose   : Copy the reviewed draft into the report fields

string Apply_AI_Report_Draft(int reportID)
{
    reportRecord = Investigation_Reports[Report_ID == reportID];
    if(reportRecord.ID == null)
    {
        return "Error: Investigation report not found.";
    }
    if(ifnull(reportRecord.AI_Draft_Summary,"").trim() == "" ||
ifnull(reportRecord.AI_Draft_Findings,"").trim() == "")
    {
        return "Error: AI draft is incomplete. Generate both sections
first.";
    }
    reportRecord.Report_Summary=reportRecord.AI_Draft_Summary;
    reportRecord.Detailed_Findings=reportRecord.AI_Draft_Findings;
    return "AI draft successfully applied to Report ID: " +
reportRecord.Report_ID;
```

```
}
```

C.3 Log_Investigation_Event

```
// Script ID : DS-06
// Location  : Function
// Purpose   : Insert an audit trail row

void Log_Investigation_Event(string eventType, string relatedRecordID,
string performedBy, string actionDetails)
{
    insert into Audit_Trail
    [
        Event_Type=eventType
        Related_Record_ID=relatedRecordID
        Performed_By=performedBy
        Event_Date_Time=zoho.currenttime
        Action_Details=actionDetails
        Added_User=zoho.loginuser
    ]
}
```

C.4 Daily_Open_Case_Review

```
// Script ID : DS-10
// Location  : Schedule - daily, Asia/Kolkata
// Purpose   : Notify officers of cases open more than 7 days

overdueDate = zoho.currentdate.addDay(-7);
overdueCases = Investigation_Cases[Opened_Date < overdueDate &&
Case_Status == "Open"];
for each  caseRec in overdueCases
{
    info "Overdue Case: " + caseRec.Case_Title;
    info "Case ID: " + caseRec.ID;
    officerEmail = caseRec.Assigned_Officer.email;
    if(officerEmail != null && officerEmail != "")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :officerEmail
            subject :"EviTrace - Overdue Investigation Case"
            message :"An investigation case assigned to you has been open
for more than 7 days.\n\nCase Title: " + caseRec.Case_Title + "\nCase ID:
" + caseRec.ID + "\n\nPlease log in to EviTrace and review this case."
        ]
        info "Notification sent to assigned officer.";
    }
    else
    {
        info "No assigned officer email found for this case.";
    }
}
```

APPENDIX D – USER MANUAL

Open the EviTrace application URL in a browser and sign in with your Zoho account. Portal users use the portal link and the credentials created for them. On a phone, install the Zoho Creator mobile app, sign in and open EviTrace. The tabs you see depend on your permission profile.

URL : https://manisham2024cserajala738.zohocreatorportal.in

Investigation Officer. Open Investigation_Cases and click Add; enter the Case Title and Opened Date, then the incident type, priority, description and assigned officer, select the CRM Organisation and CRM Contact, and submit. Use the Investigation Board Report to see cases by status and move a case through the Blueprint stages with the transition buttons. The EviTrace Command Center shows totals and charts and can open a case by Case ID.

Evidence Custodian. Open Evidence Registries and add a record with the case, evidence type, description, collector, date, storage location and initial hash, and upload the file if available. Add a Chain of Custody record for each hand-over (evidence, sender, receiver, date, reason and status) and use the COC kanban report to follow transfers by status.

Forensic Examiner. In Integrity Verification, select the evidence, enter the recorded and verification hashes and choose the result. In Forensic Examination, select the case and evidence, choose the method and write the findings. In All Evidence Registries, run Extract Evidence Text and then Generate AI Analysis from the record menu. In Investigation Reports Report, run Generate AI Draft, review it, run Apply AI Draft, edit the report and set the status to Submitted.

Auditor Reviewer. Open Approvals → Pending Tasks to approve or reject each submitted report with comments. Review Integrity Verifications, Forensic Examinations and the Audit Trail, including the timeline, without changing records.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	MANISHA M	Register No.	240701304
Project Title	EviTrace: Digital Evidence Chain-of-Custody and Investigation Tracking System	Date	10/11/2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	MANISHA M	Register No.	240701304
Department / Section	Computer Science and Engineering	Date of Viva	10/11/2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____